finalcode

March 25, 2025

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import cv2
import numpy as np
import time
import matplotlib.pyplot as plt

# Start timer
start_time = time.time()

# Load Haar cascade for face detection
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +␣
↪'haarcascade_frontalface_default.xml')

# Load YOLO model
yolo_net = cv2.dnn.readNet("yolov4-tiny.weights", "yolov4-tiny.cfg")
yolo_net.setPreferableBackend(cv2.dnn.DNN_BACKEND_DEFAULT)
yolo_net.setPreferableTarget(cv2.dnn.DNN_TARGET_CPU)


# Load YOLO class names
with open("coco.names", "r") as f:
    classes = f.read().strip().split("\n")

# Read input image
image = cv2.imread('Schools-in-Dubai-Update-C-1024x640.jpg')
if image is None:
    print("Error: Image not found or unable to load!")
    exit()
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

#   STEP 1: Detect Faces with Haar Cascade
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5,␣
↪minSize=(30, 30))
for (x, y, w, h) in faces:
    cv2.rectangle(image, (x, y), (x + w, y + h), (255, 0, 0), 2)
    cv2.putText(image, "Face", (x, y - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.5,␣
↪(255, 0, 0), 2)
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#  STEP 2: Detect Objects with YOLO
height, width = image.shape[:2]
blob = cv2.dnn.blobFromImage(image, 1/255.0, (320, 320), swapRB=True,␣
↪crop=False)

yolo_net.setInput(blob)

layer_names = yolo_net.getUnconnectedOutLayersNames()
yolo_detections = yolo_net.forward(layer_names)

# Process YOLO detections
for detection in yolo_detections:
    for obj in detection:
        scores = obj[5:]
        class_id = np.argmax(scores)
        confidence = scores[class_id]

        if confidence > 0.5:  # Confidence threshold
            center_x, center_y, box_w, box_h = (obj[0:4] * [width, height,␣
↪width, height]).astype("int")
            x = int(center_x - (box_w / 2))
            y = int(center_y - (box_h / 2))

            label = classes[class_id]  # Get object label
            color = (0, 255, 0)  # Green for YOLO objects

            # Draw YOLO detection
            cv2.rectangle(image, (x, y), (x + box_w, y + box_h), color, 2)
            cv2.putText(image, label, (x, y - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.
↪5, color, 2)

# End timer
end_time = time.time()

# Print total processing time
print(f"Time taken: {end_time - start_time:.4f} seconds")

plt.imshow(image)
plt.axis("off")
plt.show()

#  Show final image with both Haar & YOLO detections
cv2.imshow("Hybrid Detection", image)
cv2.waitKey(0)  # Wait indefinitely for a key press
cv2.destroyAllWindows()  # Close all OpenCV windows
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Time taken: 3.2397 seconds
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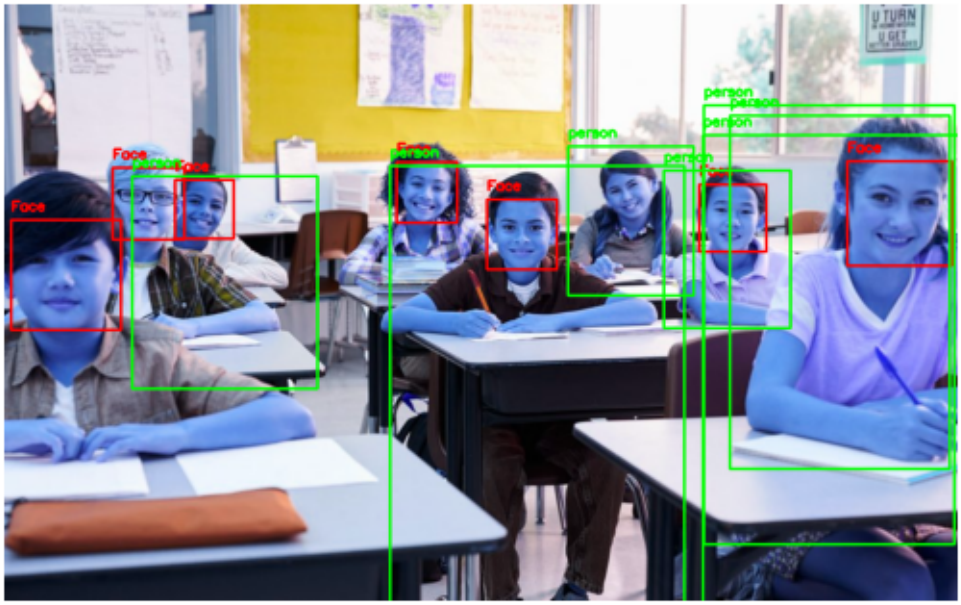
Face
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